STA 013 — Problem Set (Day 3)

Topic: Counting Rules, Permutations, Combinations & Probability | Lecture: June 24 (Professor Max)

Instructions: *Choose the best answer for each multiple-choice question. Show your work for calculation-based questions. Note: conditional probability (Q12) will NOT be on Monday's exam but is included here for early practice, per lecture.*

1. Which of the following correctly matches each term with its definition?

(i) Experimental unit (ii) Variable (iii) Discrete (iv) Continuous

A) (i) a number that changes; (ii) the object measured; (iii) measured on a number line; (iv) countable with gaps

B) (i) the object/individual measured on; (ii) a characteristic that varies across individuals; (iii) countable, with natural gaps between values; (iv) can take any value along a number line

C) (i) a category label; (ii) the population; (iii) a label; (iv) a number

D) (i) the population; (ii) the sample; (iii) qualitative; (iv) quantitative

2. A restaurant's prix fixe menu lets you choose 1 entrée (3 options), 1 side (4 options), and 1 dessert (2 options). How many distinct complete meals are possible?

A) 9

B) 18

C) 24

D) 12

→ Hint: This is a 3-stage application of the mn-rule — multiply across all stages.

3. A student has 6 different textbooks and wants to arrange ALL of them in a row on a shelf. How many distinct arrangements are possible?

A) 36

B) 6

C) 720

D) 46,656

→ Hint: Using all n objects, r = n, so this is simply n!.

4. In a race with 8 runners, medals (1st, 2nd, 3rd) are awarded in order of finish. How many different ways can the top 3 medal positions be filled?

A) 56

B) 336

C) 24

D) 512

→ Hint: Order matters here (1st ≠ 2nd ≠ 3rd), so this is a permutation: n = 8, r = 3.

5. A pizza shop offers 10 possible toppings. A customer wants to choose exactly 3 toppings, and the order they're added doesn't matter. How many different 3-topping combinations are possible?

A) 720

B) 1000

C) 30

D) 120

→ Hint: Order doesn't matter → use combinations: n = 10, r = 3.

6. A bag contains 5 marbles: 2 red and 3 blue. Two marbles are drawn WITH replacement (each marble is put back before the next draw). What is the probability that BOTH draws are blue?

A) 6/25

B) 9/25

C) 3/5

D) 6/5

→ Hint: With replacement, draws are independent — multiply P(blue) × P(blue).

7. Using the same bag (2 red, 3 blue, 5 total marbles), two marbles are now drawn WITHOUT replacement. What is the probability that AT LEAST ONE marble drawn is red?

A) 2/5

B) 3/10

C) 7/10

D) 1/10

→ Hint: Find P(no red at all) using combinations, then subtract from 1: 1 – [C(3,2)/C(5,2)].

8. Using the same bag (2 red, 3 blue), two marbles are drawn WITHOUT replacement. What is the probability of getting EXACTLY ONE red marble?

A) 3/10

B) 3/5

C) 1/5

D) 4/5

→ Hint: “Exactly 1 red” means 1 red AND 1 blue — multiply C(2,1) × C(3,1), then divide by the total C(5,2).

9. Let A = {2, 4, 6, 8} and B = {1, 2, 3, 4}. What are A ∪ B and A ∩ B?

A) A∪B = {1,2,3,4,6,8}; A∩B = {2,4}

B) A∪B = {2,4}; A∩B = {1,2,3,4,6,8}

C) A∪B = {1,2,3,4,6,8}; A∩B = {} (empty)

D) A∪B = {2,4,6,8,1,2,3,4}; A∩B = {1,3,6,8}

10. A survey of 100 people classifies them by exercise habit and health status:

	Healthy	Not Healthy	Total
Exercises	45	15	60
Does Not Exercise	20	20	40
Total	65	35	100

Let A = “Healthy” and B = “Exercises.” What is $P(A \cup B)$?

A) 0.45

B) 0.65

C) 0.80

D) 1.05

→ Hint: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. Don't forget to subtract the overlap.

11. A bowl contains 4 marbles, one each of red, blue, green, and yellow. Two marbles are drawn WITH replacement. What is the probability of drawing AT LEAST ONE red marble across the two draws?

A) 1/16

B) 1/4

C) 7/16

D) 9/16

→ Hint: Use the complement: 1 – P(no red on either draw) = $1 - (3/4)^2$. Don't confuse this with P(both red), which would be $(1/4)^2$.

12. [Practice only — NOT on Monday's exam] A class of 100 students is classified by gender and exam result:

	Pass	Fail	Total
Male	35	15	50
Female	25	25	50
Total	60	40	100

What is P(Female | Fail) — the probability a student is female, GIVEN that they failed?

A) 0.25

B) 0.40

C) 0.625

D) 0.50

→ Hint: $P(\text{Female} \mid \text{Fail}) = P(\text{Female} \cap \text{Fail}) / P(\text{Fail}) = (25/100) / (40/100)$.

Answer Key & Explanations

1. B — Experimental unit = the object/individual measured on; Variable = a characteristic that varies across individuals; Discrete = countable with natural gaps; Continuous = can take any value on a number line.

2. C — mn-rule across 3 stages: $3 \times 4 \times 2 = 24$.

3. C — Arranging all 6 distinct objects: $6P6 = 6! = 720$.

4. B — Order matters (medal positions are distinct): $8P3 = 8!/(8–3)! = 8×7×6 = 336$.

5. D — Order doesn't matter: $10C3 = 10!/(3!×7!) = 120$.

6. B — With replacement, draws are independent: P(blue)×P(blue) = (3/5)×(3/5) = 9/25.

7. C — Total ways to choose 2 from 5: C(5,2)=10. Ways with zero red (both blue): C(3,2)=3. P(at least one red) = 1 – 3/10 = 7/10.

8. B — Exactly 1 red AND 1 blue: C(2,1)×C(3,1) = 2×3 = 6. Total ways: C(5,2)=10. Probability = 6/10 = 3/5.

9. A — $A \cup B$ combines all elements from both sets (no duplicates): {1,2,3,4,6,8}. $A \cap B$ keeps only shared elements: {2,4}.

10. C — P(Healthy)=65/100, P(Exercises)=60/100, P(both)=45/100. $P(A \cup B) = 0.65+0.60–0.45 = 0.80$.

11. C — P(at least one red) = 1 – P(no red on either draw) = $1 - (3/4)^2 = 1 - 9/16 = 7/16$. (P(both red) would instead be $(1/4)^2 = 1/16$ — a different question.)

12. C — $P(Female \cap Fail) = 25/100$. P(Fail) = 40/100. P(Female|Fail) = (25/100)/(40/100) = 25/40 = 0.625.